

Computer Vision Practical Report:

IMAGE ROTATION CLOCK WISE AND ANTI-CLOCKWISE

1. AIM

To read an input image using OpenCV, perform image rotation in both clockwise and counter-clockwise directions using transformation matrices (`cv2.getRotationMatrix2D`), display the original and rotated images side-by-side using Matplotlib, and save the rotated images to disk.

2. PROCEDURE

- Load the input image and find its center point.
- Create rotation matrices for clockwise and counter-clockwise directions using `cv2.getRotationMatrix2D()`.
- Apply the rotation transformations using `cv2.warpAffine()`.
- Display the original and rotated images side-by-side using Matplotlib.
- Save the rotated images to disk using `cv2.imwrite()`

3. PROGRAM CODE

```
import os
import cv2
import matplotlib.pyplot as plt

image_path = r'C:\Users\adminuser\Desktop\Python CV\E8-IMG.png'

img_bgr = cv2.imread(image_path)

if img_bgr is None:
    print(f"Error: Could not load image from {image_path}")
    exit()

img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

clockwise_img = cv2.rotate(img_rgb, cv2.ROTATE_90_CLOCKWISE)
```

```

counter_clockwise_img = cv2.rotate(img_rgb, cv2.ROTATE_90_COUNTERCLOCKWISE)

plt.figure(figsize=(12, 6))

plt.subplot(1, 3, 1)
plt.imshow(img_rgb)
plt.title('Original')
plt.axis('off')

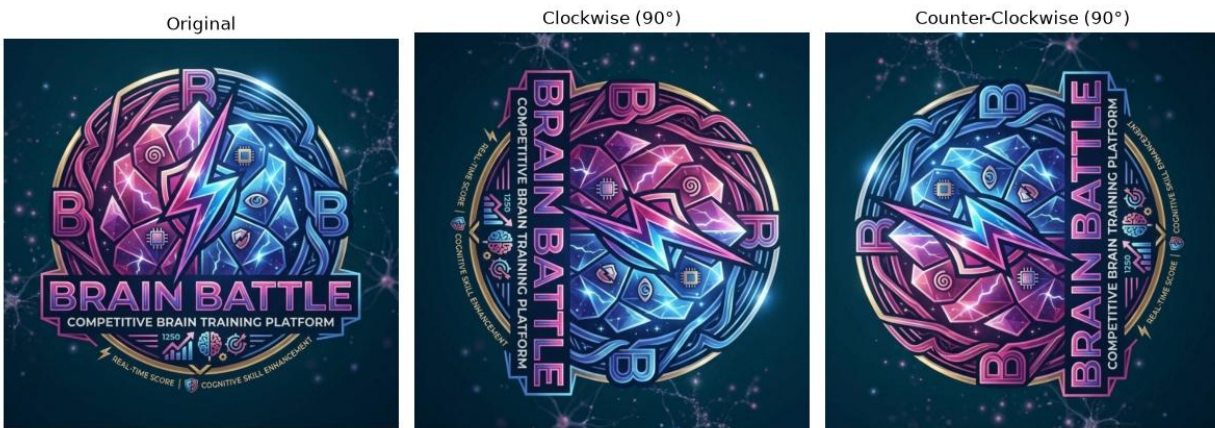
plt.subplot(1, 3, 2)
plt.imshow(clockwise_img)
plt.title('Clockwise (90°)')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(counter_clockwise_img)
plt.title('Counter-Clockwise (90°)')
plt.axis('off')

plt.tight_layout()
plt.show()

```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.